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$$\int x^3 \sqrt{x^2 + 1} dx$$

$$t = \sqrt{x^2 + 1}$$

$$\sqrt{t^2 - 1} = x$$

$$\frac{2t}{2\sqrt{t^2 - 1}} dt = dx$$

$$= \int \frac{\sqrt{t^2 - 1}^3 \cdot t}{\sqrt{t^2 - 1}} t dt$$

$$= \int \frac{(t^2 - 1)^{\frac{3}{2}} \cdot t^2}{\sqrt{t^2 - 1}} dt$$

$$= \int \frac{(t^2 - 1) \cdot \sqrt{t^2 - 1} \cdot t^2}{\sqrt{t^2 - 1}} dt$$

$$= \int (t^2 - 1) t^2 dt$$

$$= \int (t^4 - t^2) dt$$

$$= \frac{t^5}{5} - \frac{t^3}{3} + c$$

$$= \frac{(\sqrt{x^2 + 1})^5}{5} - \frac{(\sqrt{x^2 + 1})^3}{3} + c$$

References